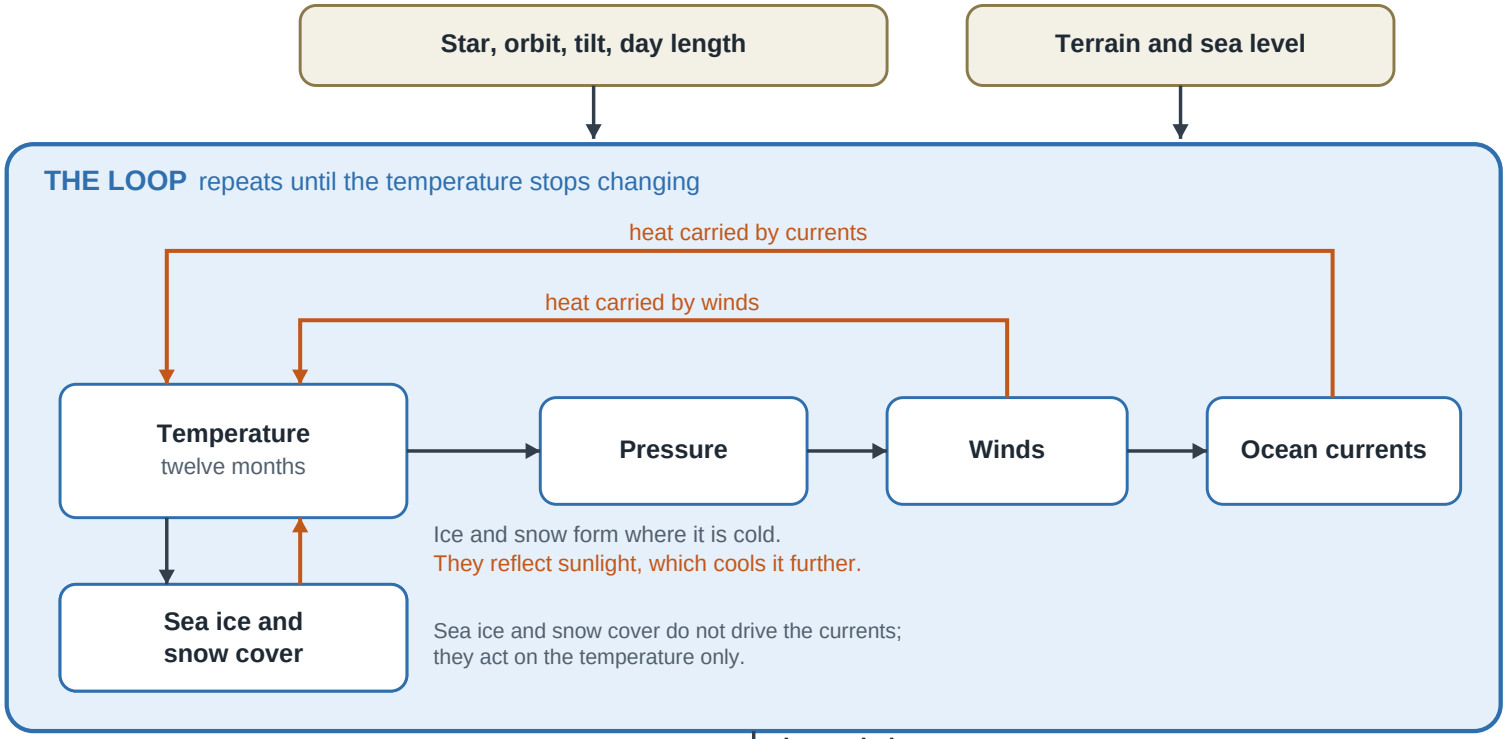
earth_sim: how the climate is worked out

The design once the biome map (roadmap step 8) is built. Drawn 14 November 2026.



Layers you can view

From the loop

- Temperature
- Pressure
- Winds
- Ocean currents
- Sea ice

Downstream

- Rainfall
- Climate zones
- Storm zones
- Biomes

Outside the loop, stepping through time

Ice sheets grow over thousands of years (step 6). Plates move over millions (step 9).

They change the terrain and the ice cover, and the loop is then run again as a snapshot.

→ feeds the next step → feeds back into the temperature

Built today (step 3, in progress): temperature, pressure and winds, worked out once in that order, with no feedback.

Planned, not built: the loop and heat carried by winds (3b), rain (4), ocean currents (5), ice (6), storm zones (7), biomes (8).